OSRIP

OSRIP: DESIGN, IMPLEMENTATION AND EVALUATION OF A BILINGUAL AI RESEARCH AND RESOURCE DECISION-SUPPORT PLATFORM FOR OMAN

A PROJECT REPORT

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ABSTRACT

OSRIP (Oman Smart Resource Intelligence Platform) is an English-Arabic academic web prototype for explaining electricity, water, food security and merchandise trade in Oman. This edition describes the software as implemented and tested: an internet-first research assistant, Al Mulhim, with three answer modes, reproducible backend calculations, structured tables, charts and an Oman-only map of 11 governorates and 63 wilayats. The 2026 NCSI Statistical Year Book supplies complete-year 2025 observations; preliminary 2026 indicators are reported only for their actual comparable periods, never as complete-year results, and missing wilayat observations are left unfilled. The repository provides one assistant persona, a welcome screen, username-and-password accounts, evidence-checked streaming answers in Arabic and English, and accessible dark and light themes. Source evidence is retained for verification while bulky source lists stay collapsed in ordinary conversations, subject to mandatory provider attribution. Automated tests run locally with scripted provider stand-ins; DeepSeek answering was verified live in October 2026, while live streaming, other external providers, public deployment and the usability study remain unverified. The evaluation covers credibility, numerical correctness, user permissions, response latency and practical decision value.

Keywords: Oman; Al Mulhim; internet research; electricity; water; food security; trade; governorates; wilayats; verified data; bilingual AI; MapLibre.

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LIST OF ABBREVIATIONS

Term	Meaning
API	Application Programming Interface
CSRF	Cross-Site Request Forgery
CSV	Comma-Separated Values
ERD	Entity Relationship Diagram
ETL	Extract Transform Load
GWh	Gigawatt-hour
MAE	Mean Absolute Error
MVP	Minimum Viable Product
NCSI	National Centre for Statistics and Information
OMR	Omani rial
OSRIP	Oman Smart Resource Intelligence Platform
RBAC	Role-Based Access Control
RMSE	Root Mean Squared Error
RTL	Right-to-left
SSE	Server-Sent Events

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CHAPTER 1 INTRODUCTION

1.1 PROJECT PURPOSE

OSRIP is a bilingual, Oman-focused academic application that connects resource indicators with independent research and practical interpretation. The four analytical axes are electricity, water, food security and merchandise trade. Users can explore reliable year-on-year comparisons, ask Al Mulhim for explanations, create structured tables or figures, and investigate shortages, surpluses, logistical constraints and relevant development questions.

The platform is a decision-support aid, not an operator of public utilities or an official statistical authority. Higher production does not automatically indicate surplus, and a rise in prices does not alone establish physical scarcity. Every interpretation must identify which observation, period, unit, geography and analytical assumption supports it.

This report documents the existing prototype, current publicly released statistics and a staged design for additions requested by the project owner. It does not fabricate field studies, deployment success, API readiness, regional percentages or completed evaluation results.

1.2 PROBLEM AND INTENDED USERS

Omani information is published through government yearbooks, regulator reports, corporate annual reports and statistical releases. Their definitions, administrative coverage and observation periods are not interchangeable. As of November 2026 the newest complete calendar year is 2025; reports published during 2026 may describe only January-September electricity or January-August trade.

Public readers may inspect available information and use the permitted guest experience; authenticated members manage their own studies. Analysts prepare approved datasets and reports, data managers review external findings, and administrators govern user groups, roles, service settings and publication. A username alone never authorizes administrative access.

The research challenge is to turn uneven source coverage into truthful, understandable comparisons while preserving evidence and not mistaking absent wilayat-level records for zero demand or production.

1.3 AIM OBJECTIVES AND QUESTIONS

The aim is to implement and assess an accessible research-and-analysis platform that helps users investigate Oman without obscuring data limits. Objectives include independent internet research, accurate annual trends, dynamic tables, exportable briefs, an Oman-centred map, secure sign-in, bilingual explanations, and a consistent single-persona interface.

RQ1 tests fidelity to primary-source definitions. RQ2 tests whether people recognize differing observation periods. RQ3 tests whether geography is correctly assigned. RQ4 evaluates whether Al Mulhim can write informative original analyses and render results progressively without producing unsupported numbers or copied source text.

Each research question has a measurable acceptance case: reproducible calculations, source traceability, valid geographic joins, appropriate permission enforcement, interface task completion, and explicit documentation of limitations.

Table 1.1 Questions and evidence

Question	Evidence required
Preserve source meaning?	Definition mapping, validation log and reviewed snapshot
Trace an indicator?	Input version, formula and matching export
Improve prediction?	Same held-out dates for baseline and candidate
Explain an action?	Rule inputs, reviewer feedback and limitations

1.4 SCOPE AND APPLICATION CHOICE

The scope includes validated published evidence, scenario assumptions and bilingual explanatory interfaces. A complete-year baseline covers 2021-2025 where the indicator exists, and the import validation rejects verified-source records dated before January 2021. A distinct preliminary 2026 layer records the actual last published month or cumulative end-date and must not be added to the full-year series.

The prototype uses React 19, TypeScript, Vite and MapLibre on the client; Python FastAPI services, PostgreSQL 17 with 31 reviewed migrations and controlled import workflows provide data, identity and server-side calculations. Existing modules were upgraded rather than rewritten from scratch; external AI and tile services remain configurable.

Excluded capabilities include direct utility operations, fabricated village or wilayat allocations, automatic approval of unverified external figures, guaranteed market-profit predictions and undeclared paid-service activation.

1.5 ELECTRICITY AND WATER DECISIONS

Electricity metrics separate gross generation, net generation, customer consumption, supply capacity and connections. The 2026 Statistical Year Book records 50,973.5 GWh of gross production for 2025, compared with 49,132.0 GWh for 2024; the 2026 updates require matched year-to-date denominators.

Water metrics distinguish production, distribution, account counts, sector geography and service-area definitions. National production was 508.1 million cubic metres in 2025 versus 520.0 million in 2024; this observation alone does not measure leaks or shortages.

Both modules should prompt users to review coverage, seasonality, unit conversion, infrastructure operation and relevant population changes before advising on future capacity or efficiency.

1.6 FOOD AND TRADE DECISIONS

Food-security analysis examines domestic production, catch and aquaculture, import exposure, commodity coverage and household price pressure. The 2025 crop figures cannot be equated with household food availability because fodder, stocks, losses and non-food uses may be included.

Trade analysis separates oil and gas exports, non-oil exports, re-exports and recorded goods imports. For January-August 2026 the NCSI reports exports above OMR 18.3 billion and imports near OMR 11.9 billion; these are preliminary partial-year figures, not annual totals.

Cross-domain recommendations address logistical capacity, water and energy intensity, trading partners and evidence gaps. They must distinguish plausible opportunity from established feasibility, and demonstrate the baseline needed to quantify any surplus.

1.7 EXPANDED RESEARCH QUESTIONS AND OBJECTIVES

Extended research questions ask whether independent live web retrieval adds useful evidence beyond existing OSRIP snapshots; whether visible writing progress improves perceived responsiveness without implying facts have already been verified; whether a single Al Mulhim identity improves continuity; and whether source disclosure on demand

can coexist with auditability and required vendor attribution. Each question requires a defined test, not promotional claims.

The corresponding objectives are three distinct assistant routes—Quick Answer, Deep Internet Research, and Data Analysis/Visualization—plus a verified research ledger, safe numeric validation, year-aware comparison, accessible cards, reports, geographic filtering and role-enforced controls.

CHAPTER 2 LITERATURE REVIEW AND BACKGROUND

2.1 RELATED STUDIES AND CRITICAL COMPARISON

Daher and Mohtar (2015) presented WEF Nexus Tool 2.0 for integrated resource-planning scenarios. The study supports connecting food choices to water and energy requirements. OSRIP adopts explicit assumptions and reproducible scenario comparisons; coefficients from another setting cannot be transferred to Oman without justification.

Taguta et al. (2022) systematically reviewed water–energy–food nexus tools and identified limitations in their accessibility and practical application. Their findings motivate testing the complete user task, from finding a statistic to checking a source and preparing a brief. They do not establish that a new dashboard will be effective for Omani users.

Al-Maskari and Ibrahim (2025) developed and assessed a Green Village model for Dar Al-Sawda, Oman, integrating solar energy, wastewater treatment and agroforestry. This local study supports examining connected development drivers. Its site-specific assumptions cannot establish nationwide benefits or justify a numerical scenario for every governorate.

Together, the studies motivate integrated reasoning, transparent inputs and practical usability. OSRIP evaluates these principles through a bilingual, source-controlled application rather than claiming a new national planning model.

2.2 OMANI STATISTICAL SOURCES

The National Centre for Statistics and Information (NCSI) is the preferred reference for national statistical series. Its Statistical Year Book 2026 (Issue 54, May 2026) presents data for 2025 and comparable earlier years. NCSI also released the Monthly Statistical Bulletin: October 2026 on 20 October; the publication title must not be confused with each table's observation end-date.

Sector companies provide a second layer: Nama Electricity Distribution, Nama Supply, Nama Power and Water Procurement and Nama Water Services have different responsibilities and reporting footprints. Nama Water Services Annual Report 2025 explicitly excludes Dhofar operations from its own service scope. Source selection must therefore record organizational coverage and prevent double counting.

Table 2.1 Source review register

Source	Use	Current boundary

Source	Use	Current boundary
NCSI electricity	Locate production and consumption series	Verified extract; coverage and revisions retained
NCSI water	Locate water production series	Production, distribution and consumption kept distinct
NCSI agriculture	Locate commodity production	Product matching still required
Oman Customs	Locate commodity trade	Detailed compatible extract not yet obtained
FAO concepts	Limit food-security interpretation	Conceptual source, not Omani observations
Forecasting textbook	Specify chronological validation	Methods source, not project results

2.3 FOOD AVAILABILITY AND INTERPRETATION

FAO distinguishes availability, access, utilization and stability; national supply does not establish household food security (Food and Agriculture Organization of the United Nations [FAO], 2008). OSRIP therefore describes commodity availability and exposure rather than declaring a complete food-security status.

Apparent availability is production plus imports minus exports, with a documented stock adjustment where known. Without compatible feed, seed, processing and loss estimates, this is not a complete food balance sheet. Missing stocks remain unknown. Local-production contribution divides production by a positive compatible apparent-availability denominator; values above 100 percent require explanation rather than automatic capping.

2.4 FORECAST EVALUATION

Forecast evaluation preserves chronology: only information available at a forecast origin enters its training data (Hyndman & Athanasopoulos, 2021). A last-observation or eligible seasonal-naive baseline and a candidate predict the same dates and horizon. Feature preparation occurs within each training window.

MAE reports errors in the indicator unit; RMSE emphasizes larger errors. Neither is an accuracy percentage. Individual errors, sample size and performance by horizon accompany aggregate results. Insufficient history disables forecasting, and a candidate that fails to improve on the baseline is not preferred merely because it is more complex.

2.5 EVIDENCE FOR THE ASSISTANT AND IDENTITY DESIGN

The single assistant persona is Al Mulhim, presented with one current portrait and chat interface; the legacy second persona and its selector have been removed from live

navigation, while earlier replies keep their stored persona under a neutral label. Progressive output streams only response segments that the model has produced and that have passed the evidence check; a prewritten answer is never animated as simulated thinking. Users retain an explicit Stop control.

Authentication uses a unique username and secret password, Argon2id hashing (Biryukov et al., 2021) and server-side permission checks. Public registration cannot grant Azzan administrator privileges: the name is reserved, the first administrator is created only by a protected command-line procedure, and later grants require an existing administrator. Single-use recovery codes and administrator-assisted reset avoid mandatory email and telephone fields, so identity recovery never becomes name-only access.

2.6 CROSS DOMAIN REASONING AND GROUNDED RESEARCH

The water-energy-food-trade nexus requires causal humility. Energy requirements for pumping, irrigation demand, crop suitability, import routes and market access can interact, but association does not prove causal impact. OSRIP must identify intermediate variables and report the information required for a defensible investment or policy study.

Internet-first research reduces dependence on stale internal snapshots, yet web search results can be contradictory and may repeat one publisher. Research must identify origin pages, distinguish independent publishers, validate units, and write a fresh synthesis rather than reproduce search snippets.

CHAPTER 3 METHODOLOGY AND SYSTEM DESIGN

3.1 APPLIED DESIGN AND EVALUATION

The methodology combines implementation audit, requirements traceability, controlled data engineering, bilingual UX design and reproducible evaluation. This edition was checked against the OSRIP repository and research workbooks on 8 November 2026; each capability is described as implemented, locally tested, live verified or proposed according to observed evidence.

Implementation proceeded from a baseline of existing routes, migrations and provider readiness to the single-persona entrance and username login, research-mode isolation, stream handling, tables and diagrams, and geographic boundaries. Final evidence and permission acceptance remains the last stage.

An auditable work log records the code revision, test environment and input dataset version for every claim. A successful static preview is not evidence of working authentication or production deployment.

3.2 ACQUISITION AND PROVENANCE

Each source is registered with publisher, document title, URL, retrieval date, licence, reporting period, unit, geographic resolution and preliminary/revised status. NCSI yearbook indicators and 2026 bulletins are separate observation vintages; an updated bulletin must not silently replace a historical series with a different definition.

For electricity, inspect NCSI gross and net generation together with the appropriate Nama reports. For water, compare NCSI production and distribution with Nama Water Services scope. Food should combine agriculture, fisheries and consumer-price evidence without treating price inflation as a direct production shortfall. Trade must retain petroleum/non-petroleum and re-export distinctions.

External search may retrieve a report in several languages or formats. The system stores a source fingerprint, extraction locator and freshness judgment, and marks unsupported or proprietary access paths as unavailable rather than assuming an API exists.

3.3 CLEANING AND AGGREGATION

Validated numeric records require a definition, comparable time boundary, standardized unit, unique geography code and reliable source locator. The import validator rejects

malformed period fields, unsupported units, duplicated natural keys and incompatible administrative scopes while retaining a row-level error report.

Published, provisional, candidate, calculated, hypothetical and synthetic values must remain separate. A whole-year 2025 total cannot be compared directly against January-September 2026. A 2026 interim comparison uses the identical January-September window in 2025 and reports its provisional nature.

When a source revises previous values, the release is stored as a new approved version, not summed with the old version. Inference below the source geography is blocked: a governorate total cannot be divided by population and shown as observed wilayat production.

Aggregation is allowed only under documented mathematical conditions. Flows may be summed across disjoint periods, rates generally require recomputation with denominators, and monetary figures may need deflation or explicit nominal labels.

3.4 ELECTRICITY ANALYTICAL SPECIFICATION

The electricity registry separates gross generation, net production, distribution, consumption, connections and installed capacity. Units such as GWh and MW are not interchangeable. The NCSI full-year gross series rises from 40,992.3 GWh in 2021 to 50,973.5 GWh in provisional 2025.

The latest validated interim comparison located by 8 November 2026 reports gross electricity production of 42,873.4 GWh in January-September 2026 versus 41,164.9 GWh in January-September 2025, a reported 4.2% increase. Net production was 41,721.4 versus 39,973.6 GWh, a reported 4.4% increase.

Analysis explores drivers such as supply assets, new connections, seasonal demand and reporting revisions. The AI may propose investigating supply adequacy, but it cannot derive actual excess capacity solely from generation growth.

3.5 WATER ANALYTICAL SPECIFICATION

Water statistics distinguish production, water delivered/distributed, sales, connections, loss studies and pumping energy. The NCSI yearbook states national production of 508.1 million cubic metres for 2025, compared with 520.0 million in 2024, while NCSI distribution totals differ in definition.

Preliminary production through September 2026 was 374.3082 million cubic metres versus 380.8105 million in the matching 2025 interval, a reported 1.7% decline. These figures cannot be interpreted as twelve-month output or used to infer wilayat-level losses.

Nama Water Services 2025 reports its organizational performance for the calendar year while excluding Dhofar. A cross-provider comparison must first reconcile area coverage, production versus distribution and account definitions before making resource-efficiency claims.

3.6 FOOD COMMODITY MATCHING

Commodity matching begins with product codes, physical units, source definitions and intended use. A wheat category from agriculture must match customs trade categories before apparent domestic availability can be estimated. Aggregated agricultural production including fodder is not a dietary supply measure.

The 2025 yearbook provides complete-year crop area and production data. Preliminary January-June 2026 fish landings total 468,407 tonnes, compared with 469,175 tonnes one year earlier. Landings must not be confused with aquaculture value, fish stocks or sustainable maximum yield.

Shortages require evidence about demand, inventories, food waste, substitution and price; a model may identify candidates for further research, but must not label an area food insecure from one national production statistic.

3.7 TRADE ANALYTICAL SPECIFICATION

Trade research records customs classifications, origin or destination, commodity, flow, period, physical quantity and nominal value. Total exports, non-oil exports, oil and gas exports, re-exports and imports are separate indicators; tariff, freight and unit-value differences prevent casual profitability claims.

An NCSI announcement dated 3 November 2026 records January-August 2026 merchandise exports above OMR 18.3 billion (+19.7% year on year), recorded imports near OMR 11.9 billion (+4.4%), and a trade surplus near OMR 6.5 billion. These provisional period totals must not be compared with full-year values.

Recommendation workflows may examine export diversification, supplier concentration, customs compliance and route resilience. They require distinct evidence on cost, capacity, demand and quality before describing a market as a commercially feasible opportunity.

Table 3.1 Indicator definitions

Indicator	Definition	Guard
Percentage change	100 × (current − previous) / previous	Comparable periods; previous nonzero
Use per subscriber	Consumption / subscriber count	Same coverage and denominator convention
Apparent availability	Production + imports − exports − stock increase	Compatible physical quantities; stock assumption shown
Local production contribution	100 × production / apparent availability	Positive denominator; not total food security
Largest partner share	100 × largest partner imports / total imports	Complete stated partner coverage
Trade balance	Export value − import value	Same currency, product and period
Mean unit value	Trade value / physical quantity	Nonzero compatible quantity; not retail price

3.8 REQUIREMENTS AND ROLES

A public visitor reads available material and uses the configured guest allowance. A registered user signs in by username and password. Analysts create structured studies, data managers validate external updates, and administrators manage users, groups, roles, permissions and service health. A username alone cannot elevate privileges.

Requirements cover source-independent Internet Research, explicit Internal Dataset Analysis, streamed final writing, chart/table output, one Al Mulhim persona, secure identity, a cinematic welcome experience, four premium sector visuals, and an Oman-only map with verified governorate and wilayat boundaries. A no-data state must be clearly distinguished from zero.

Acceptance criteria include working server sessions; denied unauthorized writes; isolated research-mode tool access; equivalent Arabic and English analytical output; real streamed chunks; accurate and reproducible calculations; correct 2026 period labelling; and accessible dark/light views on phones and desktop.

Table 3.2 Functional requirements

ID	Requirement	Acceptance evidence

ID	Requirement	Acceptance evidence
FR01	Import and validate a dataset	Invalid rows recorded; no silent deletion
FR02	Publish a version	Approved snapshot visible; previous version preserved
FR03	Explore indicators	Filters, units, period and source match
FR04	Compare compatible records	Invalid comparison rejected
FR05	Generate a report	Input versions and data mode retained
FR06	Explain a recommendation	Rule, values and next steps visible
FR07	Evaluate a forecast	Baseline and candidate share test dates
SEC01	Authorize protected actions	Denied requests leave data unchanged
UX01	Switch language	Filters and numeric values preserved
FR08	Enforce observation periods from 2021	Reject verified records before 2021; show missing coverage
FR09	Answer using approved evidence	Validated numbers and resolvable source links
FR10	Register with username/password	Unique username; ordinary role; no compulsory email/phone
FR11	Administrator group, users and role control	Denied privilege grant
UX02	Show respectful Omani portrait	Rights register and responsive review
FR12	Username/password login; no phone required	T19–T21; login tests
UX03	Accessible motion and optional 3D	T22; reduced-motion and mobile review
FR13	Separate forecasts and scenarios	T23; chronological evaluation
OPS01	Publish integrated HTTPS application	T24; external smoke test
OPS02	Visual Studio one-click local startup	T25; actual startup record
FR14	Five successful guest answers	T26 to T29; backend usage tests
FR15	Single Al Mulhim persona; optional audio	T30 and T31; speech and accessibility review
UX04	Canva asset provenance and resolution	T32; dimensions and export record
FR16	Explicit development-driver scenarios	T33; reproducible assumption record
FR17	Three AI modes with explicit tool isolation	Research mode never calls internal observation tools

ID	Requirement	Acceptance evidence
FR18	Internet-first source-verified research	T36 and T40
FR19	Oman map with verified wilayat polygons	T35; honest missing data
FR20	Well-written incremental response; qualified recommendations	Section 4.13 rubric
FR21	Reproducible cross-domain scenarios	T37
FR22	Saved research and evidence briefs	T38
FR23	2026 interim and full-year data separation	T39; reviewed updates

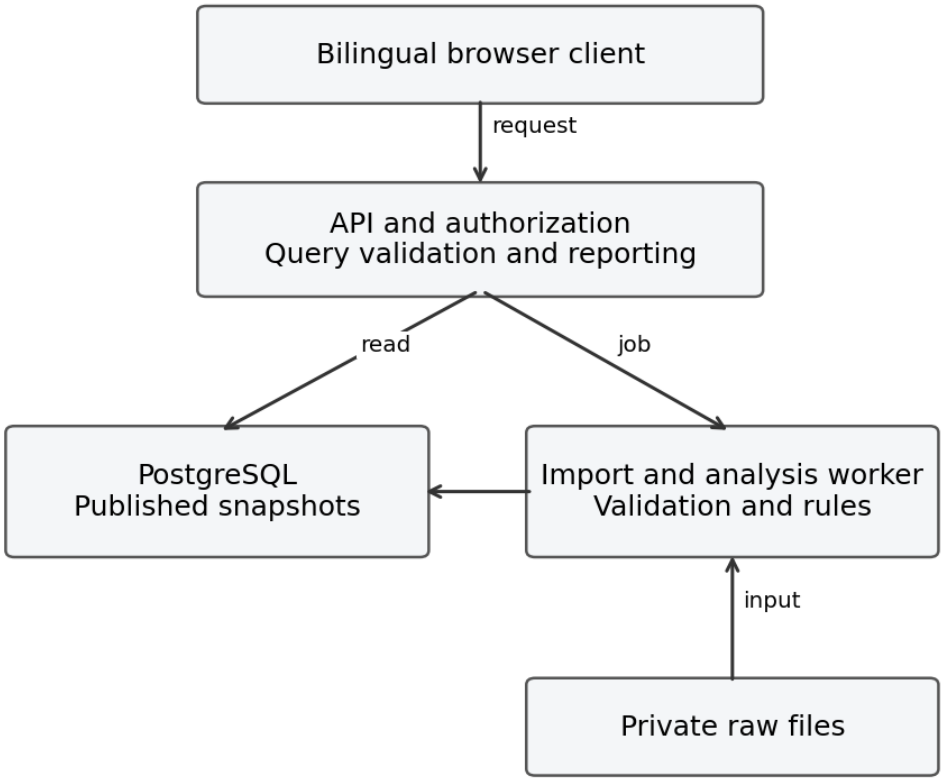
3.9 ARCHITECTURE AND INTERFACES

The frontend is React/TypeScript with a reusable component system, i18n and MapLibre; the backend is FastAPI with controlled research tools, authorization and database transactions. Adapters exist for DeepSeek, OpenAI, Gemini and Anthropic models and for DeepSeek and Brave search. Only DeepSeek is configured; its answers were verified live in October 2026, while the other providers lack working keys.

All structured outputs include original source records, published period, geography, data status, units and backend-calculated values. The client can show the conclusion first, then a table or figure. A request for source details reveals the corresponding evidence, while legally required provider attribution is retained.

Both internet modes are refused OSRIP's internal dataset tools by the backend unless the user asks for the platform's own data. Analysis mode uses selected published datasets. Quick Answer runs a short web search and, if search is unavailable, says so instead of answering from stored records.

Figure 3.1 Core data architecture



External provider failures are isolated; an inaccessible page, tile server or model does not authorize invented answers. Rate limits, cancellation, API costs, audit entries and content-security restrictions apply to every vendor.

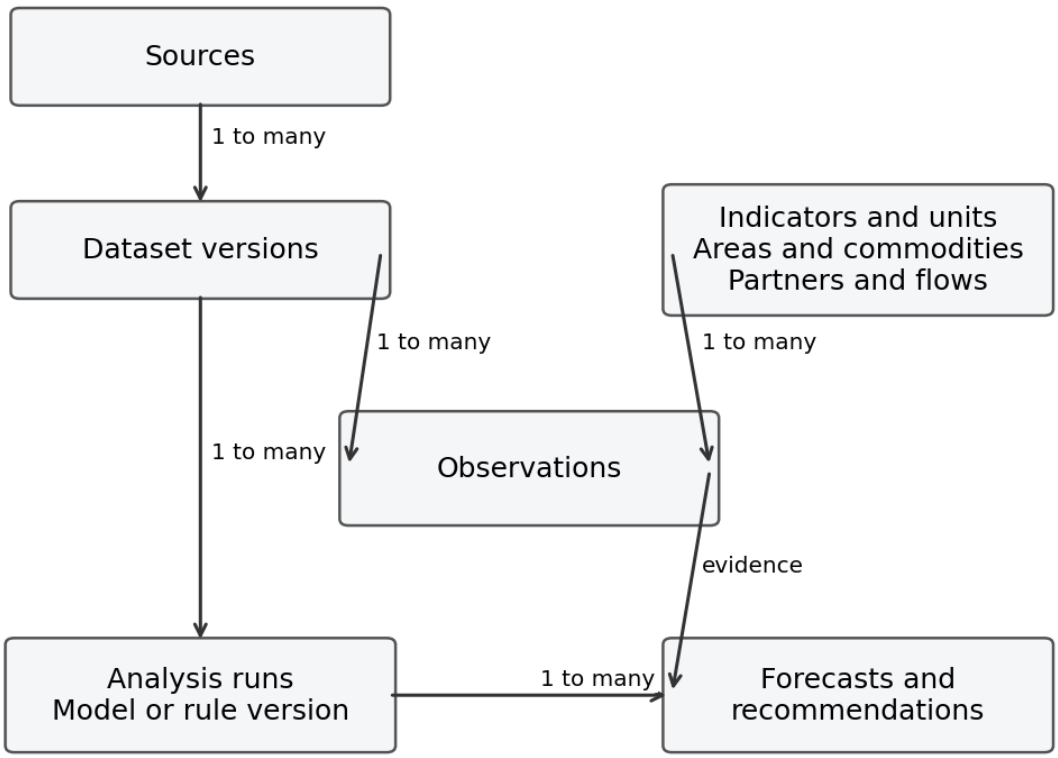
3.10 RELATIONAL DATA DESIGN

The core schema separates metadata, observed values and derived outputs. Areas, indicators, units, commodities and partners are reference entities. Sources identify publishers and definitions; dataset versions identify immutable imports. Observations reference these dimensions. Forecasts and recommendation records refer to a model or rule version and to the data version that generated them.

A natural uniqueness constraint must include the dimensions relevant to the series. For trade, omitting partner or flow would incorrectly merge different records. For national observations, an explicit national area is preferable to ambiguous null geography. Nullable dimensions require deliberate uniqueness handling; a unique constraint that permits repeated null combinations may not enforce the intended rule.

Publishing a revision does not add both versions to the current national total. Queries select a designated published snapshot or explicit version. Historic reports retain their earlier snapshot. Foreign keys prevent orphan observations, while domain checks reject impossible unit assignments. Figure 3.2 and the dictionary tables describe a logical schema; final migration syntax will depend on the implemented schema and tests.

Figure 3.2 Logical entity relationships



The diagram summarizes logical relationships. Review history is appended to recommendations and does not overwrite observations.

Table 3.3 Core entity dictionary

Entity	Purpose and key relationships
users, roles and groups	Accounts, groups and backend permissions
areas	National and regional identities with parent and validity dates
indicators and units	Definitions, dimensions, period type and aggregation rules
commodities	Product form and classification mapping
partners	Trading partner identity; no synthetic country attribution

Entity	Purpose and key relationships
sources	Publisher, URL, definition, dates and license
dataset_versions	Immutable input, checksum, review and publication status
observations	Values linked to version, indicator, area and period
trade_dimensions	Commodity, partner, flow and valuation basis
analysis_runs	Input versions, model or rule version, parameters and time
forecasts	Horizon, predicted value and supported interval
recommendations	Evidence, follow-up action and review events
reports	Frozen filters, source versions and access visibility
audit_events	Actor, action, entity, time and safe result metadata
auth_identities and sessions	Secure account identities, recovery and revoked sessions
conversations and ai_runs	Owned dialogue and reproducible model configuration
answer_evidence	References to immutable observations and calculations
groups and permissions	Memberships, role grants, scope and audit history
refresh_jobs and candidates	Revision proposals and reviewer decisions
service_health_events	Redacted API, database and refresh status

Table 3.4 Observation field dictionary

Field	Type	Validation
id	UUID	Internal key
dataset_version_id	UUID	Existing immutable version
indicator_id	UUID	Compatible indicator definition
area_id	UUID	Explicit national or regional identity
period_start / end	Date	Ordered interval with known frequency
value	Decimal or null	Null means missing; domain checks by indicator
unit_id	UUID	Same physical dimension as indicator
commodity_id	UUID or null	Required for commodity indicators
partner_id	UUID or null	Required for partner-level trade

Field	Type	Validation
flow	Controlled code	Import, domestic export or re-export where applicable
data_mode	Controlled code	Official-source or synthetic; inherited from version
quality_status	Controlled code	Accepted, warning or withheld
source_row	Text	Traceback to original file or table

3.11 API AND IMPORT WORKFLOW

Read endpoints under /api/v1 expose indicators, observations, source metadata and reports. Query parameters include area, period, commodity and data mode where relevant. A compatibility check rejects requests that mix frequencies or units, returning a stable error code and explanation without exposing database internals or credentials.

Import follows a controlled sequence: upload, map columns, validate, review errors, approve and publish. The upload is stored privately. Validation produces a summary and a downloadable error list. Approval records the reviewer and time. Publication occurs in a transaction so that a partial failure does not expose an incomplete version. Re-running a completed publication request must not duplicate observations.

A report request saves its filters and input versions before generation. The download endpoint checks ownership or report visibility. Exports include provenance and the synthetic-data label where relevant. Table 3.5 lists implemented routes; the OpenAPI description is generated from the service and exercised by integration tests.

Table 3.5 Implemented API routes

Route	Purpose	Control
GET /indicators	Definitions and compatible units	Public published metadata
GET /observations	Filtered observations and provenance	Version and mode required
POST /imports	Private file upload	Data manager
POST /imports/{id}/validate	Create validation report	Owner or permitted manager
POST /imports/{id}/publish	Publish approved snapshot	Explicit publication permission

Route	Purpose	Control
POST /analysis-runs	Run eligible rule or model	Authorized analyst
GET /recommendations	Read evidence and status	Published or permitted scope
POST /recommendations/{id}/reviews	Append review	Authorized reviewer
POST /reports	Freeze filters and generate report	Visibility and mode checks
GET /reports/{id}	Download generated report	Ownership or published access
POST /assistant/conversations/{id}/messages	Evidence-grounded answer	Session or guest allowance; read-only evidence
POST /auth/register/username	Create ordinary viewer account	Rate limited; reserved names refused
POST /auth/login	Verify password and create session	Rate limited and auditable
POST /auth/recover	Reset password with a recovery code	Single-use code; rate limited
POST /admin/groups	Create or update role group	Administrator only; audit changes

3.12 INTERPRETABLE RULES AND REVIEWER ACTIONS

Rules specify eligibility, comparison window, threshold and action. Thresholds are reviewable project assumptions, not national standards.

For example, a water-growth rule may request review when a comparable annual change exceeds a configured threshold. The card states the actual change, the threshold, the input version and the questions to check. It does not state that a leak exists. A trade rule can flag concentration in one supplier but must not convert concentration into a quantified disruption probability.

The reviewer may mark a card as under review, explained or dismissed and provide a reason. The original calculation remains immutable while the review history is appended. This distinction is important: changing a review status does not change the statistical observation. The evaluation will inspect whether explanations accurately represent their inputs and whether reviewers understand the difference between evidence and action.

Initial rules cover four review situations:

R01 Demand growth: verify comparable periods and thresholds, then review coverage, subscribers and seasonality.

R02 Supplier concentration: require complete partner quantities, then investigate supply alternatives and contract context.

R03 Value and quantity divergence: compare compatible trade records, then review unit values and product composition.

R04 Data quality: withhold comparisons affected by missing periods, conflicting units or failed reconciliation.

3.13 BILINGUAL INTERFACE AND MAPS

The language selector changes interface direction, number formatting and explanatory text without changing observed values. The welcome screen, rebuilt from the owner's design, names OSRIP, shows the four sector images, waits for the visitor and offers Skip and Replay, without motion under reduced-motion settings. Sign-in asks only for a username and password.

The Oman-only map uses MapLibre, with MapTiler styles once a key is configured (MapTiler is untested live) and OpenFreeMap with open hillshading meanwhile. Musandam, all 11 governorates and 63 wilayats use simplified OpenStreetMap boundaries (OpenStreetMap contributors, 2026); a wilayat shows a value only where independently observed, so all currently show No data.

A compact collapsible sidebar groups the axes, research and analysis, geographic, report, account and administration pages. Selected year, metric and geography should remain consistent across the map, table, chart and generated report.

3.14 SECURITY AND RESEARCH ETHICS

Public data does not remove the obligation to protect accounts, uploads and unpublished analysis. The implementation uses Argon2id password hashing, single-use recovery codes, bounded sessions, backend RBAC and audit events. The first administrator is created only by a protected command that sets a temporary password at a hidden prompt, to be changed at first sign-in; later grants need an existing administrator, the name Azzan is reserved, and no sample credential exists in the repository. Sign-in is throttled after five failures and queries are parameterized. These controls are tested locally; no independent security assessment has been made.

Uploads are untrusted input. The service limits file size and supported formats, checks schema and parses data without executing embedded content. Report downloads are permission checked. Logs record actions and outcomes but exclude passwords, tokens and unnecessary personal information. A September 2026 restore drill recovered a known dataset version into a separate database (Section 3.18).

If usability research involves participants, the team will obtain the college's required clearance before recruitment. Participation will be voluntary and task records will use codes rather than names. The report will describe the actual recruitment and consent process only after it occurs. Signature fields and approval references remain blank until completed by the responsible people; this document is not an approval certificate.

The proposed usability sample is 12 adults aged 18 or above: eight students and four lecturers recruited voluntarily at the college after approval. One session of approximately 30–40 minutes uses test accounts and public or synthetic data. No audio or video recording is planned. Coded task notes and questionnaire responses are stored separately from consent records; proposed retention is 12 months after assessment, subject to college policy. Recruitment must avoid assessment pressure and permit withdrawal before anonymized aggregation.

3.15 DELIVERY PLAN AND OWNERSHIP

Work is divided into four streams: frontend, backend, data and analytics, and database with quality assurance. Ownership assigns responsibility for a deliverable but not exclusive knowledge. Every student should review a component outside their main stream and explain how a published value reaches the dashboard. A common repository, issue list and definition of done will coordinate the work.

The revised schedule reserves weeks one to four for indicator coverage and provider feasibility, weeks five to eight for schema, ordinary username/password login and the first import-to-chart slice, and weeks nine to twelve for the AI assistant, geographic view, scenarios and visual refinement. Weeks thirteen to sixteen cover group-based authorization, data-refresh approval, bilingual accessibility, real provider checks and deployment. Optional voice and elaborate 3D assets are deferred before source quality or access control is weakened.

The highest risks are incompatible food and trade definitions, limited historical series and scope expansion. The responses are an early commodity-selection gate, a simple baseline or disabled forecast, and exclusion of unrelated features. Hosting costs will be recorded

from actual choices; no fixed budget or commercial return is claimed here. A release is ready only when its known limitations and test evidence are recorded with its code version.

Table 3.7 Delivery schedule

Weeks	Deliverable	Review gate
1–2	Problem, users and platform choice	Supervisor scope confirmation
3–4	2022 onward source mapping	Compatible input definitions
5–6	Schema and interface prototype	Migration and screen review
7–8	Verified accounts and data path	Permission and import tests
9–10	Domain views and Omani imagery	Independent calculation fixtures
11–12	Grounded AI and optional forecast	Baseline comparison plan
13–14	AI security and integrated tests	Recorded defect resolution
15–16	Usability, report and demonstration	Evidence audit and final review

3.16 GEOGRAPHIC REFERENCE DATA AND SPATIAL INTERPRETATION

The geographic registry includes the 11 governorates and 63 wilayats used by current Omani publications, including Musandam. Names, codes, versions and parent-child relations are recorded independently of polygon geometry. Old boundary datasets must be checked for missing or renamed administrative areas.

The map draws simplified OpenStreetMap boundaries (ODbL) for orientation only; authoritative geometry is needed before legal or official use. Terrain and satellite basemaps are not administrative boundaries. The mainland, Musandam and coastal features must be framed consistently.

Area-level indicators are joined through stable geographic codes and their original source resolution. If only a national or governorate amount is published, the map shows it at that level and keeps wilayats in an explicit no-data state.

Wilayat centroids can support keyboard selection but do not locate actual facilities. Relief, elevation, coastlines and administrative boundaries come from versioned geographic layers; they do not substitute for measured electricity, water, food or trade values. A regional investment suggestion needs site-level logistics, supply, environmental and commercial evidence. The map is a decision-support view, not proof of project viability.

3.17 REPOSITORY STRUCTURE AND IMPLEMENTATION SEQUENCE

The repository separates the web interface (frontend), service code (backend/app), reviewed migrations (backend/alembic), controlled fixtures and project documentation (docs). Frontend feature folders group domain pages with their presentation components, while reusable controls belong in a shared interface layer. An API client centralizes request handling and error interpretation. Business calculations are independently testable backend functions rather than chart code.

The backend separates request validation, authorization, application services and persistence. Route handlers translate a request into a service operation and return a documented response, so the same rule can be explained and tested without a browser.

Implementation began with one vertical slice: a controlled input file, a validation result, an approved database version, an indicator endpoint and a page that displays that indicator, followed by a matching report. The pattern was then repeated for other domains, exposing definition and integration problems early.

Dependency lockfiles, migrations and secret-free configuration examples are committed; real keys stay in an untracked local environment file.

The implemented stack is React with TypeScript for the interface, FastAPI with Python for validation and analysis, and PostgreSQL for durable records. The frontend consumes one typed API contract. Model, search, speech and email adapters are separately configured, so an unavailable provider does not disable ordinary statistical browsing.

3.18 DEPLOYMENT CONFIGURATION AND OPERATIONAL RECOVERY

The initial deployment can use a single application environment with a separate database service, provided it preserves the boundaries described in the design. Public web assets and backend requests should use secure transport in a deployed environment. Database credentials, session secrets and environment-specific addresses belong in protected configuration. Example configuration files should contain placeholders and explain required variables without exposing real credentials.

Publishing requires a real hosting environment, a reachable HTTPS endpoint and successful access from another device. The requested osrip.project.com address is a subdomain of project.com and requires the domain owner’s authorization and DNS control. Renaming an HTML file cannot create that address. OAuth redirects, cookie

settings, CORS and API URLs must match the actual deployed origin; no public address is claimed in advance.

Development, evaluation and demonstration data should be distinguishable. A reset script for synthetic fixtures must never target an official published dataset by default. The deployment process should apply migrations, confirm database connectivity and run a small health check before exposing the new version. The exact hosting choice remains an implementation decision; this report does not claim that hosting has been purchased or deployment has occurred.

Recovery planning should address both software and data changes. A previous application version may be restored only if it remains compatible with the database schema. Destructive migrations require special care, because restoring code alone may not recover removed data. Before a migration that changes stored observations, the team should document a recovery procedure and test it in a disposable environment.

Backup usefulness is demonstrated through restoration, not merely by a backup file. A drill on 25 September 2026 restored a pg_dump copy of the development database into a separate database; table counts, observation checksums and a known dataset version matched, and published observations stayed immutable. The drill predates later migrations; point-in-time recovery, off-site copies and scheduled backups remain untested.

3.19 EXTENDED DATABASE AND API CONTRACTS

The API contract distinguishes research answers, streamed events, source evidence and dataset observations. Each response includes stable IDs, status, language, statistics, optional tables, chart specifications and the originating research job; the interface must never parse numbers from explanatory text to calculate indicators.

Provider readiness is a server-side state with unavailable, configured, tested and failed values. Search credentials and usage budgets remain backend-only. A test response from a mock provider is not evidence of real OpenAI, Brave or MapTiler access.

The application exposes reviewed research candidates separately from published data and restricts publication to authorized roles. Source identities persist even if ordinary answer screens collapse their details.

AI-service credentials remain exclusively on the backend. Responses must not expose raw prompts containing internal instructions, private imports or credentials. An assistant timeout returns a recoverable status while the statistics dashboard stays usable.

Conversations are kept until the user deletes one or all of them, the owner's chosen policy; college retention rules must be confirmed before participants use the system.

3.20 USERNAME-AND-PASSWORD LOGIN AND SECURE ACCOUNT RECOVERY

Accounts use a username and password, without compulsory email, phone, SMS or WhatsApp. Public registration creates viewer accounts with names unique in any letter case; passwords are hashed with Argon2id. Session cookies are HttpOnly and SameSite=Strict, sessions are time-limited, and state-changing requests carry a CSRF token.

The administrator Azzan is created only with the ensure-admin command, which sets a temporary password typed at a hidden prompt that must be changed at first sign-in; typing the name never grants rights, and the command refuses to promote a self-registered account. Users change passwords after signing in; recovery uses single-use codes or an administrator-assisted reset, never a name-only reset.

A design preview does not establish working authentication. Browser tests exercise real sign-in, case-insensitive duplicate usernames, wrong passwords, recovery codes and refused administration for an ordinary account; backend tests cover throttling and the idle limit.

The role model separates guest visitors from viewer, analyst, data manager and administrator accounts. A group may contain several users and roles; effective permissions are derived and checked at each protected endpoint, and role or group changes create audit events.

The owner sets the administrator's first secret privately on the target environment, never with an example password in source files, fixtures or reports, and rotates any previously shared credential. Later groups and grants are made by an administrator and logged.

Email uses a development outbox until SMTP is configured; the optional SMS and WhatsApp routes report themselves unavailable without a provider. None is required for normal sign-in, and none has been verified with a live provider.

3.21 WINDOWS STARTUP AND VISUAL STUDIO INTEGRATION

The requested development experience is a single Start action after a documented initial setup. Microsoft Visual Studio and Visual Studio Code are different environments. Visual Studio supports multiple startup projects, but the concrete configuration must match the

existing solution and installed workloads (Microsoft, n.d.). A VS Code launch file alone does not establish Visual Studio compatibility.

The React, FastAPI and PostgreSQL architecture is retained. OSRIP.sln and the VS Code tasks run tools/start_osrip.py, a standard-library launcher that checks prerequisites, starts the portable PostgreSQL cluster, applies forward-only migrations, starts the backend and frontend, waits for readiness and then opens the local website.

Initial setup records the supported Python and Node versions, dependency installation, environment-file creation and database migrations. Missing prerequisites produce an actionable message. An ordinary run does not install privileged software, reset the database or change machine-wide security policies.

The launcher handles quoted Windows paths, configured ports, timeouts and partial startup failures, and never terminates a process it did not start. A browser opens only after the frontend responds and the backend readiness check confirms database connectivity and current migrations.

Stopping a session ends only the processes it started, and re-running reuses a running OSRIP instead of duplicating it. Visual Studio's Run started the launcher on 8 October 2026, and the launcher was ready in 13 seconds on 11 October; local execution is not internet publication.

3.22 SINGLE-PERSONA AL MULHIM AND STREAMED CONVERSATION

OSRIP presents one assistant persona, Al Mulhim (الملهم), from assistant profile version 4. The Ilham selector and imagery have been removed, while earlier replies keep their stored persona under a neutral label. The portrait is labelled as a generated image of a fictional Omani man, never an actual human or official adviser.

Answers appear progressively as the model writes: each sentence is released only after passing the same evidence check as the final answer and is sent as a server-sent event (WHATWG, 2026). Searching, reading, analysing, writing and checking states are shown, with Stop and Ask again. This is tested with stand-ins; live DeepSeek streaming is unverified.

Optional audio uses the male Omani voice ar-OM-AbdullahNeural once Azure AI Speech is configured (it is not yet), otherwise a male browser voice where installed. Text remains fully capable; microphone failure never blocks analysis, tables or reports.

Arabic and English share conversation state and the guest allowance. Users explicitly choose language; tool results preserve dates, units and chart filters. The default assistant panel prioritizes a clean narrative, generated tables and charts without visible reference lists. A “Show sources” control, or a request in words, reveals the stored evidence when the user needs to inspect or export it.

Speech recognition and synthesis are separate adapters whose bilingual quality requires testing. Permission denial, unsupported browsers and provider outages leave text usable. Decorative audio is not a functioning assistant.

Raw audio retention is off by default; any exception requires a purpose, notice and bounded retention. Keys remain server-side. Cancellation and interrupted streams must not create another question.

3.23 FIVE ANSWER GUEST ALLOWANCE

A guest may receive five successful assistant answers in total across Arabic and English, text and voice. The fifth answer is delivered completely. The sixth request requires sign-in or registration before the backend calls the answer provider. The public statistics pages remain available without an account.

A successful answer is a non-empty completed response that meets the response contract and is stored for retrieval. Failures and rejected requests do not consume allowance. A stored answer replayed under the same request ID does not incur another charge.

The server stores usage against a random anonymous-session cookie. Refreshes, tabs and language changes preserve the count. Clearing cookies or changing devices can create a new anonymous identity; perfect cross-device enforcement is not claimed.

A transaction reserves an available slot before calling the provider. Success consumes it and genuine failure releases it. Idempotency keys prevent duplicate reservations. Concurrent requests cannot exceed the remaining allowance. Reconcile expired reservations with completed responses before releasing them.

Responses include remaining usage and an authentication-required state; the backend enforces both. Rate limits complement the allowance without equating a shared IP address with one person. Intrusive fingerprinting is excluded.

Authenticated users have a separate fair-use policy. Linking guest history requires explicit authenticated consent and proof of session control. An unverified email address or profile entry cannot merge accounts or grant elevated permissions.

Table 3.8 Guest and voice service records

Record	Key content
guest_session	Random session identity, expiry, completed-answer count and policy version
assistant_request	Session or account, idempotency key, reservation state, outcome and response reference
speech_sessions	Transient request reference, selected language, permission and cancellation state
asset_versions	Original file, production tool, export dimensions, provenance and optimized derivatives

The guest_session and assistant_request records exist in the migrated schema and are covered by integration tests; speech_sessions and asset_versions remain proposed. Private prompts, recordings and credentials are excluded from usage counters.

3.24 VISUAL SYSTEM, ACCESSIBLE THEMES AND ASSET PROVENANCE

The design language is a restrained charcoal, warm peach and amber glassmorphism system, with ivory-based light mode. Reusable tokens define contrast, focus, surface opacity, typography and hover states. The appearance must not obscure geographic labels, numerical values or data-quality warnings.

The existing introductory artwork and four separately generated resource cards can be reused with recorded creation dates and rights. Generated visuals are illustrative and do not constitute documentary photographs of actual dams, plants or ports. The newest single-persona portrait replaces the obsolete assistant screen everywhere.

An accessible evaluation checks Arabic shaping, RTL navigation, tab order, legible chart legends, responsive table controls and reduced-motion support. No dashboard layout or intro animation is allowed to make authentication, data provenance or statistical clarity harder to use.

Canva is an optional visual-production tool when its account permissions and export capabilities are verified (Canva, n.d.). Any exported assets belong in durable project storage with licensing and dimensions documented, not temporary download links. UI behavior itself remains implemented in the web application.

Respect prefers-reduced-motion, contrast, responsive image sizes, explicit labels and keyboard paths in both themes.

Animation libraries such as Motion may support discreet interactions if compatible (Motion, n.d.). Assistant activity states and route transitions reflect actual events. A visually striking glass card cannot conceal missing units, provenance, analysis limits or

service errors. Test mobile performance and ensure background blur never reduces form or chart legibility.

3.25 THREE AI MODES, INTERNET-FIRST ROUTING AND CONTEXT

Al Mulhim supports three modes: Quick Answer for low-latency questions, Deep Internet Research for external evidence and original synthesis, and Data Analysis/Visualization for selected user-approved datasets and reproducible computation. The backend refuses internal OSRIP dataset tools in both internet modes, and a failed web search is reported rather than silently replaced with stored platform records.

Each request carries language, question, permitted tools, geography and desired output type. Search results are evaluated as untrusted data, not instructions. Follow-up context may retain legitimate user selections, but switching modes updates the tool permissions explicitly.

Speed optimization combines model routing, genuine streaming, concurrent independent searches where safe, cancellation and caching of retrievable public facts. Faster output must not bypass numerical verification, evidence boundaries, permissions or the distinction between current web findings and archived data.

3.26 EXTERNAL RESEARCH AND SOURCE REVIEW WORKFLOW

Deep Internet Research performs targeted search, retrieves original pages and documents, checks dates, measures and definitions, and writes its own analytical synthesis rather than copying snippets. High-value Omani primary sources include NCSI and the appropriate Nama operating company; independent confirmation means separate original publishers, not several AI models repeating the same source.

A research ledger holds retrieval timestamps, source locations, exact statistical values, confidence flags and licensing obligations. In ordinary chat, large source lists and raw evidence identifiers are hidden; explicit requests reveal the evidence tied to the exact answer. Required provider citations or attributions remain visible when terms demand them.

If one statement cannot be verified, the system keeps the otherwise supported analysis and omits or qualifies that statement with a note. Answer withheld is used only when too little supported content remains.

An authorized refresh process detects potential updates and stages them in a candidate queue. The data manager maps dimensions, validates units and periods, checks duplicates

and approves a new immutable version. Only approval updates dashboard charts, wilayat indicators and published reports. Failed or unreviewed refreshes cannot silently replace good data; users can see the published-as-of date.

External content is untrusted: the system must resist prompt injection, private-network fetching, unsafe redirects and oversized downloads. OpenAI or another language-model adapter uses secret server-side configuration and explicit tool scopes. Track request budgets, errors and cache age; a scheduled refresh needs ownership, review logs and a tested recovery path.

3.27 RESEARCH DRIVERS ACROSS THE PROJECT DOMAINS

Electricity research tests demand and production patterns, installed supply, connection changes, reliability, weather exposure and network limitations. A production increase alone does not establish spare capacity; the system states which load and asset evidence would be required.

Water research checks sources, distribution, service coverage, demand, conservation and energy costs. Agriculture connects crop types to irrigation and supply chains. Fisheries examines landings and market value without inferring stock sustainability from a single catch change.

Trade research evaluates commodity codes, market partners, logistics, concentration, capacity and quality standards. Every recommendation identifies what is known, what is inferred and what evidence would establish feasibility.

Trade analysis separates values, quantities, domestic exports and re-exports. Freight, standards, tariffs and global prices are candidate influences. Supplier diversification may mitigate concentration, but productive surplus requires buyers, logistics capacity and viable margins. Higher nominal trade value does not prove higher physical volume.

For each domain, record the measured gap or surplus, relevant drivers, missing evidence, candidate actions and trade-offs. Cross-domain scenarios need compatible coefficients: for example, crop expansion may alter pumping electricity, water use and trade. Do not extrapolate a local assumption to nationwide trade without evidence.

3.28 RECOMMENDATION CARDS AND EVIDENCE STRENGTH

Recommendations are structured as issue, observation, possible driver, option, required data, limitation and suggested next investigation. A no-action monitoring alternative is

included when the evidence is insufficient. The assistant does not declare particular regions or firms profitable based on aggregate trade growth.

A shortage hypothesis requires a documented demand or requirement baseline, not only declining output. A surplus hypothesis needs verified available supply relative to demand or export capability; national production is not transferable to individual wilayats.

Responses are written in natural Arabic or English prose and may render a relevant table, decision matrix or chart. Evidence is retained behind the interface and available through an explicit citation request rather than a 100-item source dump.

Present alternative actions, a do-nothing/monitor option and trade-offs. Rankings require user-selected criteria with transparent weights; unknown inputs cannot receive favorable default scores. Human reviewers retain responsibility for data publication, procurement decisions and permissions.

3.29 COMPARISON AND SCENARIO WORKSPACE

The Analysis/Data/Scenarios workspace combines one validated dataset, its filters, its reproducible calculations, a sortable table and matching charts. Users choose axis, indicator, compatible calendar years, governorate or wilayat, and the permitted data mode. The map and report must preserve those choices.

Forecasts require adequate continuous history and chronological validation. Scenarios are separate user-stated assumptions and display sensitivity to coefficients; they cannot be presented as observed trends or statistically validated future values.

Cross-domain cases may connect crop irrigation requirements, pumping electricity, transport costs and commodity demand only when coefficients, spatial scope and reference years are recorded. Missing parameters yield a research question rather than a fabricated output.

Model forecasts have a separate record containing training coverage, data version, evaluation split, baseline and error measures. Eligibility depends on frequency, sample size and target horizon. Four or five annual observations alone do not justify a complex model. Future driver values must be known or explicitly forecast at the evaluation origin to avoid leakage. Chronological evaluation follows the methodology described by Hyndman and Athanasopoulos (2021).

3.30 SAVED RESEARCH ALERTS AND DOWNLOADABLE BRIEFS

Users can save axis choices, research, filter sets and scenario assumptions against immutable versions. Refreshing makes a new reviewed result without overwriting historical reports. Theme or language preferences never alter measured values. Personal and group reports require explicit access checks.

A source watcher may flag revised releases, stale indicators or thresholds. Each job records source version, schedule, outcome and reviewer decision. Deduplication and recovery prevent repeated notices. Alerts are investigation prompts, not claims of shortages. Notifications remain optional until delivery is configured and tested.

Reports collect question, geography, period, methods, charts, tables, scenario inputs and recommendations. PDF, Excel and CSV exports preserve units, version identifiers and missing-value flags. The main UI can hide reference lists, but academic reports include the source register and synthetic-data disclosures.

The source library separates approved observations, unreviewed candidates and retired versions. Evidence remains accessible for charts and claims. A worker command runs due source checks when invoked, but no operating-system schedule is set up, so scheduled retrieval is not yet operational.

3.31 EXTENDED RECORDS AND SERVICE CONTRACTS

Research jobs record mode, language, prompt category, provider, allowed tools, completion state, latency and cost estimate. Each evidence item retains a publisher, document locator and provenance status; it does not become a published observation without a reviewed import.

User permissions are enforced by the server and apply independently of the conversation UI. Stream events convey search progress, verified output segments, table/chart specifications and final completion, without revealing private credentials.

Administrative dashboards distinguish actual observed uptime and expenditure from simulated or unknown values. Dataset versions, source identities and report exports remain reproducible.

Voice remains an optional adapter with transcript review and explicit microphone permission. AI responses use readable text, tables and charts; the user may open Evidence Details rather than seeing citations in every reply. Raw audio storage is off by default, and unavailable speech never disables the text assistant.

3.32 FEASIBILITY COSTS AND SERVICE GOVERNANCE

Feasibility must be proven through live authentication, complete import-to-report workflows, independent research tests, correct statistical comparisons and responsive UI checks. The existing repository includes implementation components, but the presence of an adapter does not establish operational API keys or a publicly deployed service.

Recurring costs may include model tokens, search calls, image generation, map tiles, database, storage and hosting. A customer-facing subscription is not a developer API subscription. No service may be purchased or enabled for billing without separate explicit approval and a configured spending cap.

Service Health reports actual backend, database and model readiness, rejected external updates, latency and quota. Its metrics must not be imitated with static positive badges.

A useful release delivers evidence-backed improvements for electricity, water, food security and trade, while keeping incomplete governance, geography or external integration tasks visibly pending.

CHAPTER 4 WORKED EXAMPLES AND EVALUATION PLAN

4.1 ELECTRICITY AND WATER EXAMPLES

The electricity fixture uses synthetic annual generation of 48,000 GWh and 50,100 GWh. The absolute increase is 2,100 GWh and the percentage increase is 4.375 percent, displayed as 4.38 percent after rounding. This is a generation example. It cannot support a statement that household consumption rose by the same percentage. The complete years and identical units are explicit assumptions of the fixture.

The water fixture uses synthetic consumption of 330 and 343 million cubic metres. The increase is 13 million cubic metres and the relative change is approximately 3.94 percent. No source breakdown or subscriber count accompanies the fixture, so per-subscriber use and changes by source remain unavailable. A correct interface must keep those fields unavailable rather than derive them from the total.

Both cases are tested at the calculation layer with these fixture values. The stored values retain their precision, while display rounding occurs only at presentation. If the previous value is zero, the percentage-change field explains that the comparison is undefined. If the previous value is missing, the result is missing rather than zero growth.

Table 4.1 Synthetic fixture inputs

Fixture	Earlier value	Later value	Unit
Electricity generation	48,000	50,100	GWh
Water consumption	330	343	Million m³

4.2 FOOD BALANCE EXAMPLE

Consider a synthetic commodity with production of 100,000 tonnes, imports of 80,000 tonnes and exports of 20,000 tonnes in one year. Under an explicitly zero stock-change assumption, apparent availability is 160,000 tonnes. Production divided by this balance is 62.5 percent. Gross imports divided by the balance are 50 percent. These ratios do not sum to 100 percent because exports are a separate component.

For an illustrative supplier breakdown, the 80,000 tonnes of imports are allocated to Partner A, B and C as 48,000, 20,000 and 12,000 tonnes. Their shares are 60, 25 and 15

percent. The largest-partner share is therefore 60 percent. These partner labels are fictional and should never be presented as claims about real countries.

The example supports a review question: should alternative suppliers be investigated? It cannot establish disruption probability or justify a particular stockpile. The calculation tests verify the 160,000-tonne availability and the 62.5 percent production share, and that all components refer to the same product and year. The zero stock-change assumption must travel with the exported calculation.

4.3 TRADE AND INTERPRETATION EXAMPLE

The same synthetic commodity can be used for a trade calculation without changing its physical flows. Assign imports of 80,000 tonnes a value of OMR 40 million and exports of 20,000 tonnes a value of OMR 12 million. The commodity balance is minus OMR 28 million. The mean import unit value is OMR 500 per tonne and the mean export unit value is OMR 600 per tonne.

The difference between these unit values is not a profit margin. Import and export baskets can differ in quality, processing and valuation basis. The prototype will therefore avoid a recommendation such as buy at 500 and sell at 600. Instead it will display the values and request investigation of product comparability and actual transaction costs before any commercial conclusion.

This fixture tests whether the food and trade pages use consistent quantities but distinct interpretations. The food page examines apparent availability; the trade page examines monetary flows. If a user selects a partner subset, both the chart and denominator label must update. A subset must not silently replace the national total in a recommendation that still claims national coverage.

Table 4.2 Expected arithmetic outputs

Calculation	Expected result	Interpretation
Electricity growth	4.375% before rounding	Synthetic generation change
Water growth	3.939393…%	Synthetic consumption change
Food apparent availability	160,000 tonnes	Assumes zero stock change
Production contribution	62.5%	Not a complete food-security measure
Gross import contribution	50%	Does not complement production ratio
Largest partner share	60%	Fictional Partner A

Calculation	Expected result	Interpretation
Commodity trade balance	−OMR 28 million	Not business profit or loss
Import unit value	OMR 500/tonne	Not retail price
Export unit value	OMR 600/tonne	Not a resale margin

4.4 FUNCTIONAL AND INTEGRATION TESTING

Functional tests use both synthetic fixtures and approved statistical releases, but synthetic values never appear as official data. The workbook import test checks that January-September 2026 electricity figures are staged only beside the matching January-September 2025 interval and never as a full year, while 2025 remains the latest completed calendar year.

Identity tests verify username/password login, denial of unauthorized group actions and the guest allowance, including a race for the last slot. Assistant tests confirm that only Al Mulhim appears, persona controls are absent, text is shown while it is written and before the stored reply, and the internet modes cannot query OSRIP dataset tools.

Evidence outcomes record test case, date, version, provider configuration, actual output and whether the result was mocked, locally tested or live verified. On 8 November 2026 the backend suite passed 1,039 tests, and the frontend type check, lint, contrast check, build and 17 unit tests passed. The full-stack browser suite recorded 199 passed, 14 failed and 3 skipped: six failures were Talk-page assertions still expecting the removed persona choice, one was a genuine one-pixel clipping of the message box in a very short window, and seven were timeouts while the computer slept. After the assertions were updated and the message box given a scroll margin, all 14 passed when re-run. All model and search providers were scripted stand-ins. No test is reported as passed without observed execution.

Table 4.3 Planned acceptance tests

ID	Test input or action	Expected outcome
T01	Previous value is zero	Undefined percentage explained
T02	Value absent in source	Missing retained in chart and export
T03	Mix GWh with MW	Comparison rejected
T04	Upload duplicate natural key	Duplicate reported; no double count

ID	Test input or action	Expected outcome
T05	Select overlapping source versions	One explicit snapshot used
T06	Public user requests publish	Denied; no data change
T07	Synthetic mode report	Visible synthetic label and metadata
T08	Switch Arabic to English	Same filters and numbers
T09	Trade partner subtotal	Partial coverage clearly labeled
T10	Generate forecast with inadequate history	Forecast withheld with reason
T11	New import fails during publication	Previous published snapshot preserved
T12	Export the commodity fixture	160,000 tonnes and stock assumption retained
T13	Import pre-2021 observed records	Excluded from release dataset
T14	Ask for absent official value	Unavailable; no invented statistic
T15	Ask AI to ignore permissions	No unauthorized retrieval or action
T16	Reuse a recovery code or session token	Rejected after one valid use
T17	Register with administrator role field	Role escalation denied
T18	Inspect portrait on mobile	Readable layout and documented rights
T19	Register and log in by username/password	Ordinary role; no mandatory email or phone
T20	Request account recovery	Ownership check, expiry and one-time token
T21	Attempt unauthorized group grant	Denied; no role modification
T22	Reduced motion and mobile navigation	2D fallback; focus and filters preserved
T23	Insufficient series or hypothetical inputs	Forecast withheld; scenario labelled
T24	Access from external device after deployment	HTTPS tested, only if actually deployed

4.5 CHRONOLOGICAL FORECASTING EXPERIMENT

Forecast evaluation begins by selecting a series with a clear frequency, stable definition and sufficient history for the proposed horizon. The selection should be justified before comparing models. If the available history is too short or changes definition repeatedly, descriptive analysis is an acceptable outcome. A model should not be fitted simply to populate a forecast panel.

The evaluation plan should specify the target, horizon, training period, validation windows and final holdout period. At every forecast origin, only information available by that date may be used. Any scaling, feature selection or missing-value treatment must be fitted within the training window. A revised historical release should be identified as such; using final revised data may answer a different question from simulating decisions made with the original releases.

The baseline should be simple and appropriate to the series. A last-observation forecast can be compared with a seasonal-naive forecast when a seasonal cycle is supported. The candidate and baseline must predict the same target dates using the same information boundary. Error tables should retain individual forecast values, actual observations and errors, allowing aggregate measures to be checked.

MAE should be interpreted in the target unit, while RMSE gives greater weight to large errors. These measures do not provide a general accuracy percentage. Results should be discussed by horizon and against the baseline, with sample size and unusual periods disclosed. Forecast intervals may be shown only when their construction and evaluation are documented. If the candidate fails to improve on the baseline, the report should retain that finding and recommend the simpler method for the prototype.

4.6 TASK BASED USABILITY STUDY

The proposed usability study asks whether participants can interpret the application correctly and complete a small set of representative tasks. Recruitment begins only after the required college clearance. The team should describe eligibility, consent, the planned session length and the information recorded. Participation should be voluntary, and participants should be able to stop without having to justify their decision.

Tasks should describe goals without explaining which button to press. One task can ask a participant to locate a selected indicator for a stated period and identify its unit. Another can ask the participant to explain the difference between official-source and synthetic data. A third can require opening the source details, reproducing a percentage change and exporting the selected view. A final task can ask what a recommendation does and does not establish.

The facilitator should record completion, critical errors, requests for help and observed misunderstandings. Assistance must be noted because completing a task with guidance is different from completing it independently. Comments should be recorded using participant codes, with consent records stored separately. A small convenience sample can

reveal interface problems but cannot establish that all Omani users will understand the system.

Analysis should connect observations to design changes. If several participants interpret generation as consumption, the response may involve clearer labels and indicator definitions rather than a cosmetic colour change. After revisions, selected tasks can be repeated to check whether the problem persists. The final report should explain recruitment, language coverage, task results and limitations. No participant quotations, satisfaction scores or completion times are claimed in the current edition.

4.7 PERFORMANCE EXPERIMENT AND INTERPRETATION

Performance tests compare first response time, total answer latency, search retrieval duration, stream chunk timing and report finalization under realistic network conditions. Measurements are separated for a short question, long independent research and numeric table/chart generation.

A fast initial token is not a reliable final answer. The evaluation captures numbers of searches, tool calls and completed claims, and tests user cancellation, provider timeouts and delayed external pages.

Load-testing and monitoring measure actual resources rather than vendor benchmarks. A September 2026 run on the development laptop gave single-client warm p95 latencies below 0.6 s, but 4.7 s with ten concurrent clients and one worker, above the two-second target. With scripted providers, Quick Answer took 2.0 s against 5.0 s for Deep Internet Research with concurrent searches (5.8 s sequential). Live DeepSeek answers took about 7 to 10 s in October 2026 without streaming; time to the first streamed sentence has not been measured live. Costs are recorded alongside latency.

The quality gate requires unchanged arithmetic, appropriate uncertainty, authorized access and retrievable evidence after performance optimization.

4.8 VERIFIED ANNUAL AND 2026 INTERIM ELECTRICITY/WATER

The latest complete-year electricity and water data in the NCSI Statistical Year Book 2026 refer to 2025. Gross generation was 50,973.5 GWh in 2025 versus 49,132.0 GWh in 2024 (+3.7% by calculation). Water production was 508.1 versus 520.0 million cubic metres (-2.3%). These are yearly comparisons, not 2026 annual results.

Table 4.5 Electricity and water: January-September 2025/2026

Indicator	Jan-Sep 2025	Jan-Sep 2026	Unit
Gross electricity generation	41,164.9	42,873.4	GWh
Net electricity generation	39,973.6	41,721.4	GWh
Water production	380,810.5	374,308.2	Thousand m³

The newest preliminary 2026 update located by 8 November covers January-September: 42,873.4 GWh gross electricity generation (+4.2% year on year), 41,721.4 GWh net generation (+4.4%), and 374.3082 million cubic metres of water production (-1.7%). The corresponding 2025 partial-year values and period labels are printed in Table 4.5.

NCSI reported the preliminary figures through the Oman News Agency on 7 November 2026. Provider-specific Nama annual reports describe company operations and may exclude particular areas; they are complementary evidence rather than directly additive national totals.

4.9 VERIFIED FOOD PRODUCTION AND FISHERIES

Annual 2025 NCSI agricultural figures belong to the completed-year baseline. The published crop area was approximately 326 thousand feddans; the yearbook includes 4,085,864 tonnes of crops including fodder. This total is not equivalent to food consumed by households and does not by itself indicate domestic self-sufficiency.

Table 4.6 Food production and fisheries observations

Indicator	Period	Value	Unit
Total crops including fodder	2025	4,085,864	Tonnes
Vegetables	2025	1,531,838	Tonnes
Dates	2025	405,997	Tonnes
Wheat	2025	10,194	Tonnes
Capture fish landings	Jan–Jun 2026	468,407	Tonnes
Fish value including aquaculture	Jan–Jun 2026	320.795	Million OMR

The newest verified fisheries interim total in this review is 468,407 tonnes landed during January-June 2026, compared with 469,175 tonnes for January-June 2025 (-0.2%). Reported fisheries value including aquaculture was OMR 320.795 million (+1.5%). The different scope of physical landings and monetary value must remain explicit.

Price pressure adds an independent dimension: NCSI reported an 8.03% annual rise in the food and non-alcoholic beverages CPI category for September 2026, but this is an inflation rate for one month, not a cumulative food-production figure or proof of shortage.

4.10 FOOD SELF SUFFICIENCY STATEMENTS

Officially reported commodity self-sufficiency ratios may support an illustrative sector snapshot, but the stated reporting basis, definition and actual observation year must be checked before placing them in an annual comparison. A publication date in 2026 is not necessarily the underlying crop-year date.

Table 4.7 Self sufficiency ratios as reported in May

Commodity	Reported ratio
Fish	146%
Dates	99%
Milk	96%
Eggs	95%
Vegetables	79%
White meat	62%
Red meat	45%
Fruit excluding dates	24%

Ratios for fish, dates, milk, eggs, vegetables and meat reflect different production/consumption frameworks. They cannot be added, averaged or transferred to wilayats. Table 4.7 retains the exact publisher-reported values as contextual statements rather than a comparable time series.

Food security additionally depends on affordability, access, quality, stocks, supply-chain stability and water availability. AI analysis should propose specific further measurements before interpreting a self-sufficiency ratio as resilience.

4.11 2025 ANNUAL AND 2026 INTERIM MERCHANDISE TRADE

The latest full-year merchandise baseline available in this review is 2025, while the NCSI announcement of 3 November 2026 contains provisional January-August 2026 totals. It reports exports greater than OMR 18.3 billion (+19.7%), imports around OMR 11.9 billion (+4.4%) and an approximately OMR 6.5 billion trade surplus.

Table 4.8 Merchandise trade: January-August 2025/2026

Indicator	Jan-Aug 2025 (approx.)	Jan-Aug 2026 (approx.)
Total exports	15,300	>18,300
Oil and gas exports	9,800	12,100
Non-oil exports	4,400	4,900

Indicator	Jan-Aug 2025 (approx.)	Jan-Aug 2026 (approx.)
Reexports	1,100	1,300
Recorded imports	11,400	11,900

Within the same 2026 period, oil and gas exports were approximately OMR 12.1 billion, non-oil exports OMR 4.9 billion, and re-exports OMR 1.3 billion. Figures are rounded in the source; neither totals nor a trade balance should be recomputed from separately rounded or differently defined components.

The UAE, Saudi Arabia and India are among named non-oil export destinations in the NCSI release, but destination shares require the full denominator and commodity scope. A change in value may reflect prices, quantity, product composition or reporting changes.

OSRIP therefore keeps a source-controlled annual series for 2021-2025 and a separately labelled 2026 year-to-date panel. On 8 November 2026 most of the owner's 2022-2025 annual values were already published in OSRIP with identical figures, and the January-June 2026 fisheries totals were published; the 2021 baseline and the 2026 electricity, water and trade figures were staged as validated imports awaiting a data manager's approval. Source metadata are maintained for audit even when normal AI conversations do not show a long bibliography.

4.12 GUEST ALLOWANCE AND BILINGUAL VOICE EVALUATION

The five-answer guest allowance is validated on the server by integration tests across languages, modes, failures, cancellations and replays with the single Al Mulhim persona. An interrupted or failed answer does not consume a completed-answer slot, and user-facing counters agree with backend state.

Concurrent requests for the last guest slot use a transactional reservation, and a test races two requests for it. Retried identifiers do not duplicate charges or responses. Account creation uses a chosen username and password, granting the viewer role without email or mobile verification.

Optional speech evaluation measures permission handling, transcript accuracy and interruption under Arabic and English. The visible assistant avatar is the approved new single-persona design; no legacy portrait or second-persona switch remains in active navigation.

Accessibility tasks check readable streamed response chunks, Stop and Retry controls, screen-reader announcements and correct RTL layout, with keyboard operation and a static alternative under reduced motion.

Table 4.9 Extended release acceptance

Test	Action	Expected evidence
T25	Run the Visual Studio startup profile	Browser opens after API, frontend and database readiness
T26	Send five successful guest questions	Exactly five answers; one shared bilingual allowance
T27	Send a sixth question	Authentication required before provider invocation
T28	Retry failures and duplicate request IDs	No lost allowance and no duplicate charge
T29	Race two requests for the last slot	At most one new answer reservation succeeds
T30	Deny microphone access or cancel recording	Text remains usable; recording stops
T31	Inspect Arabic and English speech responses	Transcript, numeric agreement and visible citations
T32	Inspect Canva and 4K claims	Actual provenance and dimensions; transformations disclosed
T33	Run a scenario with missing coefficients	Assumptions visible; no fabricated causal prediction

4.13 EVALUATION OF RESEARCH QUALITY AND RECOMMENDATIONS

A research-quality benchmark should use matched Arabic and English questions across Quick Answer, Deep Internet Research and Data Analysis/Visualization. Cases must include recent 2026 developments, conflicting providers, missing wilayat statistics, unit mismatch, and Arabic phrases for trade and resource questions.

The primary comparison is independent web mode against selected internal-data mode. No web search may silently call the internal observation database; the web-only test must show which tool categories were permitted and which were excluded.

Reviewers assess grounded claim accuracy, numeric reproducibility, synthesis quality, source freshness, actionability, and proper period labelling. When a claim is unverified, only the unsupported part is withdrawn; a useful supported response is retained.

Streaming tests confirm that text appears while the answer is being written, before the stored reply exists, and that Stop ends it without charge; chunk timing with a live provider is not yet recorded. Image generation is evaluated separately from trustworthy charts; artwork must not serve as statistical evidence.

Proposed gates require deterministic calculations to match their fixtures; no invented sources, regional values or percentage denominators; no unsupported savings; and correct refusal on every missing-data case. Login and group permissions must resist privilege escalation, and external updates must not publish without review. These are test criteria, not achievements; failures need a rerun record.

4.14 INTEGRATED ACCEPTANCE AND USER COMPREHENSION

End-to-end acceptance includes the cinematic introduction, unique username/password login, one Al Mulhim persona, genuine progressive research writing, a clean answer layout, secure evidence disclosure, consistent data tables, approved dataset publication and accessible map behavior.

Users should be able to choose a sector, ask for the latest available results, distinguish 2025 complete-year figures from 2026 preliminary partial-year observations, and understand why a wilayat lacks a value. The interface must not interpret missing data as zero.

Only executed tests may be labeled passed. The final appendix preserves test IDs and reviewer evidence and leaves unavailable external keys, hosted services and unfinished geographic layers explicitly unresolved.

CHAPTER 5 CONCLUSIONS AND RECOMMENDATIONS

5.1 CONTRIBUTION LIMITATIONS AND DECISION VALUE

OSRIP aims to make Omani resource and trade information understandable through verified computation, independent internet research and accessible bilingual reporting. The principal contribution is not a model that always has an answer, but a workflow that distinguishes facts, calculations, interpretations, scenarios and missing evidence.

The recent NCSI and Nama publications demonstrate why time period and provider scope must be visible: 2025 is the latest complete calendar year, while 2026 electricity, water, fisheries and merchandise trade updates end in different months. The same discipline applies to geography; national totals cannot create invented wilayat percentages.

Automated test records, provider-readiness logs and the boundary data licence now accompany the code, but live evidence beyond DeepSeek answering, deployment records and evaluated usability outcomes do not yet exist. This report does not substitute for those records.

5.2 DELIVERY SEQUENCE FOR AL MULHIM AND OSRIP

Priorities (1) to (5) are implemented and tested locally: preserving working code and data, the new opening and username/password accounts, retiring the second persona, evidence-checked streamed answers and isolating internet research from internal observations. Priority (6), validating tables, charts and provider attribution with live providers, remains open.

Subsequent work should obtain authoritative boundaries, configure MapTiler terrain without unauthorized billing, approve the staged 2021-2026 statistics, measure live streaming latency and improve reports. Every stage must pass regression tests and be pushed to a review branch rather than merged automatically.

Requirements and statistical evidence were reviewed on 8 November 2026. Publication in 2026 does not imply that any complete-year 2026 figures exist; project capabilities remain labelled implemented, tested, configured or proposed according to observed evidence.

REFERENCES

Al-Maskari, S., & Ibrahim, B. (2025). Integrated water-energy-food system for rural climate adaptation: The Green Village case in Oman. Project literature review; verify publication details before final submission.

Biryukov, A., Dinu, D., Khovratovich, D., & Josefsson, S. (2021). Argon2 memory-hard function for password hashing and proof-of-work applications. IETF RFC 9106. https://www.rfc-editor.org/rfc/rfc9106

Brave Software. (2026). Brave Search API documentation and pricing. https://api.search.brave.com/

Canva. (n.d.). Canva Developers documentation. https://www.canva.dev/docs/

Daher, B. T., & Mohtar, R. H. (2015). Water-energy-food nexus tool 2.0: guiding integrative resource planning. Water International. Verify citation details against the publisher record.

DeepSeek. (2026). DeepSeek API documentation: chat completions, streaming and search. https://api-docs.deepseek.com/

Food and Agriculture Organization of the United Nations. (2008). An introduction to the basic concepts of food security. https://www.fao.org/

Food and Agriculture Organization of the United Nations. (n.d.). Water-energy-food nexus. https://www.fao.org/land-water/water/watergovernance/waterfoodenergynexus/

Google AI for Developers. (2026). Gemini API: search grounding and tool-use documentation. https://ai.google.dev/gemini-api/docs/

Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and practice (3rd ed.). OTexts. https://otexts.com/fpp3/

Lodderstedt, T., Bradley, J., Labunets, A., & Fett, D. (2025). Best current practice for OAuth 2.0 security. IETF RFC 9700. https://www.rfc-editor.org/rfc/rfc9700

MapLibre. (2026). MapLibre GL JS documentation. https://maplibre.org/maplibre-gl-js/docs/

MapTiler. (2026). MapTiler Cloud documentation and plan information. https://docs.maptiler.com/

Microsoft. (n.d.). Set multiple startup projects. Visual Studio documentation. https://learn.microsoft.com/en-us/visualstudio/ide/how-to-set-multiple-startup-projects

Motion. (n.d.). Motion for React documentation. https://motion.dev/docs/react

Nama Electricity Distribution. (2026). Annual report 2025. https://www.distribution.nama.om/financial-information

Nama Power and Water Procurement. (2026). Annual report 2025. https://www.omanpwp.om/annual-report

Nama Water Services. (2026). Annual report 2025: Operational coverage excludes Dhofar Governorate. https://nws.nama.om/

National Centre for Statistics and Information. (2026a). Statistical Year Book 2026, Issue 54 (data 2025). https://api.ncsi.gov.om/uploads/pdfs/statistical_year_book_2026___issue_54_1777963604.pdf

National Centre for Statistics and Information. (2026b, October 20). Monthly Statistical Bulletin: October 2026. https://www.ncsi.gov.om/publications

National Centre for Statistics and Information. (2026c, November 3). Merchandise exports exceed OMR 18.3 billion by August 2026. https://www.ncsi.gov.om/newsPage/newsDetail?id=420

National Centre for Statistics and Information. (2026d). Consumer Price Index: Foods and non-alcoholic beverages, September 2026. https://data.gov.om/

Oman Daily / Oman News Agency. (2026, November 7). Preliminary electricity and water production through September 2026. https://www.omandaily.om/article/1214358

OpenAI. (2026). Responses API: web search, tool use and streaming documentation. https://platform.openai.com/docs/

OpenStreetMap contributors. (2026). Administrative boundaries of Oman, retrieved through the Overpass API on 1 November 2026. Open Database Licence 1.0. https://www.openstreetmap.org/copyright

OWASP Foundation. (2026). Authentication cheat sheet. https://cheatsheetseries.owasp.org/cheatsheets/Authentication_Cheat_Sheet.html

OWASP Foundation. (2026). LLM prompt injection prevention cheat sheet. https://cheatsheetseries.owasp.org/cheatsheets/LLM_Prompt_Injection_Prevention_Cheat_Sheet.html

Royal Oman Police, Directorate General of Customs. (2026). Oman Customs: trade and tariff services. https://www.customs.gov.om/

Taguta, C., Senzanje, A., Kiala, Z., Malota, M., & Mabhaudhi, T. (2022). Water-energy-food nexus tools in theory and practice: a systematic review. Frontiers in Water, 4, 837316. https://doi.org/10.3389/frwa.2022.837316

WHATWG. (2026). HTML Living Standard: Server-sent events. https://html.spec.whatwg.org/multipage/server-sent-events.html

APPENDIX A PROJECT RISK REGISTER

Table A.1 Risk register

Risk	Owner	Response
Incompatible commodity definitions	Data lead	Select compatible commodities before modeling
Insufficient time series	Analytics lead	Use descriptive analysis or baseline
Web format not accepted	Team and supervisor	Confirm application form before platform commitment
Scope expansion	Team coordinator	Defer tourism, navigation and extra sectors
Unauthorized data publication	Backend and QA	Server-side checks and audit trail
Synthetic data mistaken for official	All members	Mode separation in storage, display and export
Missing final evidence	QA lead	Weekly evidence review; no fabricated results
Arabic layout defects	Frontend lead	Test RTL tables, units, navigation and reports
Unsupported AI response	AI and QA lead	Evidence validation and release gate
Provider or email unavailable	Backend lead	Visible unavailable state; tested alternatives
Portrait rights unclear	Frontend lead	Use approved licensed or labelled illustrative asset
Authentication or recovery abuse	Backend lead	Rate limits, secure recovery and audited denials
Unreviewed external data refresh	Data lead	Approval gate and version history
Misleading wilayat percentage	GIS lead	Supported geography or no-data label

The risk register assigns mitigation owners; unresolved risks remain visible in the release decision.

APPENDIX B EVALUATION AND SUBMISSION RECORDS

Use this form for each executed test, including normal login, two themes, verified wilayat mapping, approved data refresh, AI evidence access, group permissions and admin service health. Keep participant identities separate from test records and follow approved retention procedures.

Test ID and requirement: _____

Code version and input snapshot: _____

Environment and date: _____

Steps performed: _____

Expected output: _____

Actual output: _____

Pass or fail and evidence reference: _____

Reviewer and unresolved issue: _____

Usability session record

Participant code: _____

Task and language: _____

Completion without help: _____

Time and critical error: _____

Observed difficulty: _____

Participant comment: _____

Proposed interface correction: _____

Submission evidence includes the final source revision, migrations, setup instructions, test and provider-readiness logs, Oman boundary licensing, security review, academic permissions and similarity report. No real administrator password, AI token, private email credential or participant identity belongs in this booklet. Record deferred functionality and failed checks honestly.

APPENDIX C 2026 DATA CURRENCY AND PROVIDER AUDIT

Evidence cut-off: 8 November 2026. The 2026 Statistical Year Book (Issue 54) was published in May 2026 but describes completed calendar-year 2025 observations. As of this revision, no complete calendar-year 2026 production or trade total exists. The separate preliminary updates below must retain their actual year-to-date end dates.

Electricity: NCSI preliminary gross production was 42,873.4 GWh for January-September 2026 versus 41,164.9 GWh for January-September 2025 (+4.2% reported); net output was 41,721.4 versus 39,973.6 GWh (+4.4%). These records cannot be extended to October without a published observation.

Water: preliminary production reached 374,308.2 thousand m³ through September 2026 versus 380,810.5 thousand m³ one year earlier (-1.7%). Nama Water Services Annual Report 2025 covers its own operating area and excludes Dhofar; it is not interchangeable with the national NCSI production series.

Food security: capture-fish landings totalled 468,407 tonnes through June 2026 versus 469,175 tonnes for the matching period (-0.2%). The September 2026 food-and-nonalcoholic-beverages consumer-price inflation rate was 8.03% year on year. Neither series directly quantifies food shortages at wilayat level.

Trade: NCSI reported preliminary exports above OMR 18.3 billion, imports near OMR 11.9 billion and an approximately OMR 6.5 billion merchandise surplus for January-August 2026. Rounding and trade-definition differences mean the published surplus should be retained rather than reconstructed from rounded figures.

Geographic coverage: the official reference register has 11 governorates and 63 wilayats, but not every axis has published wilayat-level values. A map must display the original geographic resolution and mark unavailable wilayat data explicitly. Rates require a compatible numerator and denominator.

Provenance and release gate: retain original URLs, publisher, retrieval timestamp, observation end date, units, provisional status and version for each indicator. The public AI narrative may collapse long source lists, but reviewers must be able to inspect evidence and required third-party attribution must not be removed.

Release status: in OSRIP the 2026 electricity, water and trade figures await a data manager's approval.

APPENDIX D VISUAL ASSET REGISTER

The images below remain generated concepts, not verified facilities or native-4K outputs. The fictional Omani professional and the electricity, water and trade images support the original design direction. The revised interface also uses the supplied glassy dashboard references for styling cues; licensed theme assets, login screenshots, GIS relief layers and food imagery must be documented when produced.



Fictional Omani professional
1024 × 1536 pixels



Electricity concept
1536 × 1024 pixels



Water concept
1536 × 1024 pixels



Trade concept
1536 × 1024 pixels

Figure D.1 Existing project visual concepts